Distributed Systems

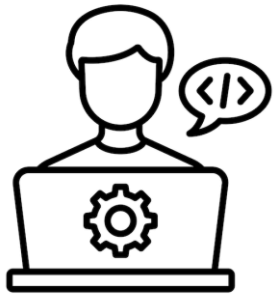
Distributed Systems/Platforms

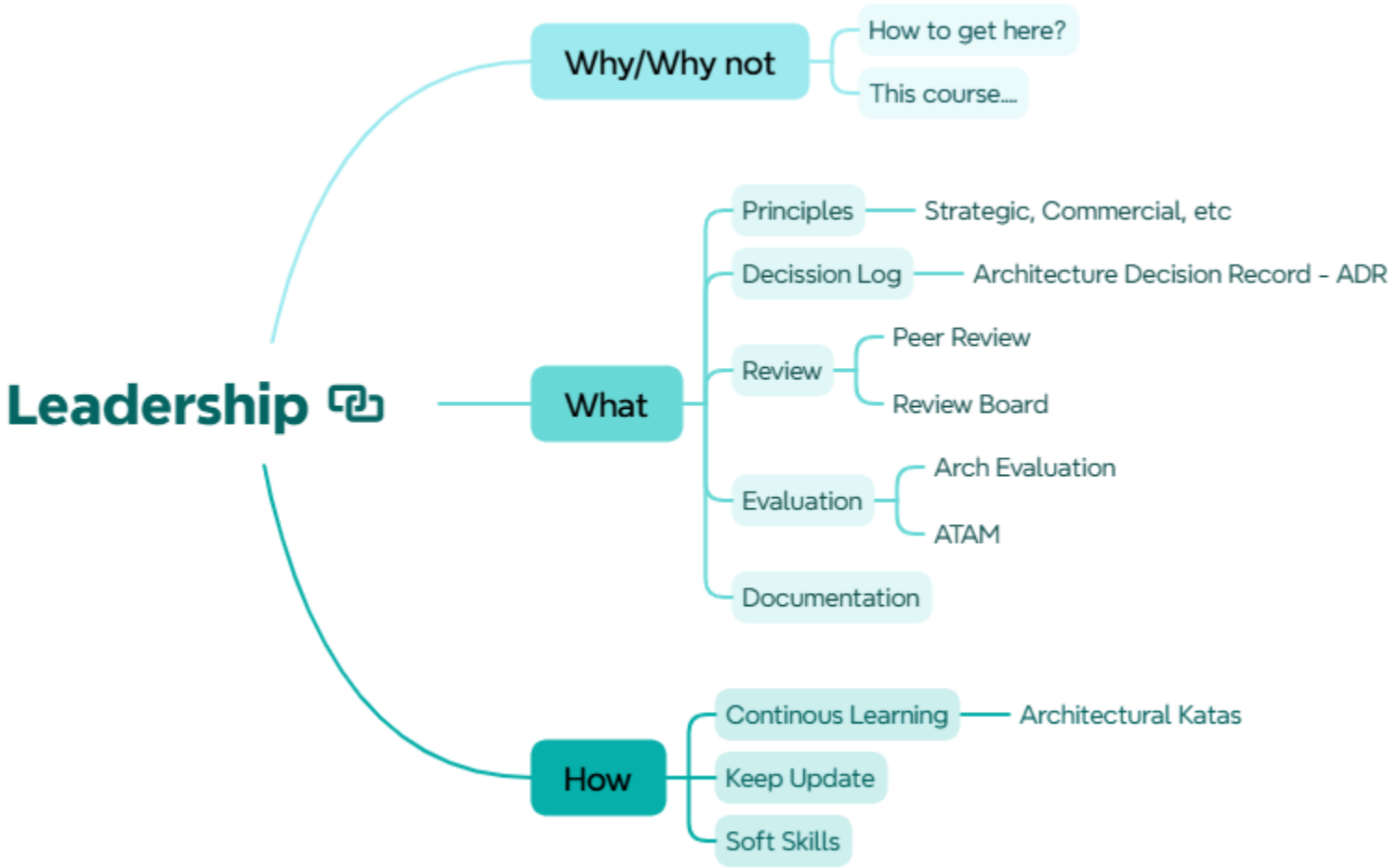


Universidad de La Sabana

FACULTAD DE INGENIERÍA

Agenda




Leadership
Why/Why not
How to get here?
This course....
What
Principles
Strategic, Commercial, etc
Decission Log
Architecture Decision Record - ADR
Review
Peer Review
Review Board
Evaluation
Arch Evaluation
ATAM
Documentation
How
Continous Learning
Architectural Katas
Keep Update
Soft Skills

Universidad de La Sabana
FACULTAD DE INGENIERIA

How to get here ?

So far, you have…

- Define Quality Attributes to apply on architecture
- How to prioritize them and discover additional requirements
- Understand the need of Arch Patterns and Styles
- List steps for ADD, DDD and Assemblage to apply on pattern to create an architecture
- Go deep on defined Arch Styles and Patterns: Microservices and Sharding

Then, you miss…

- How to register our activity and decisions ?
- How to evaluate our work? Who?
- How to create an environment that people follow you?

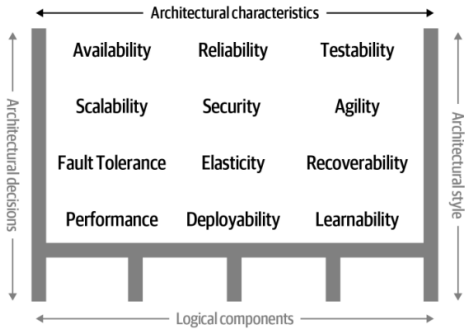
Expectations

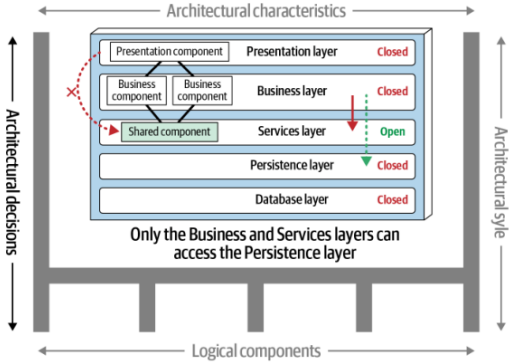
- Make architecture decisions
- Continually analyze the architecture
- Ensure compliance with decisions
- Keep current with latest trends
- Understand diverse technologies, frameworks, platforms, and environments
- Know the business domain
- Lead a team and possess interpersonal skills
- Understand and navigate organizational politics

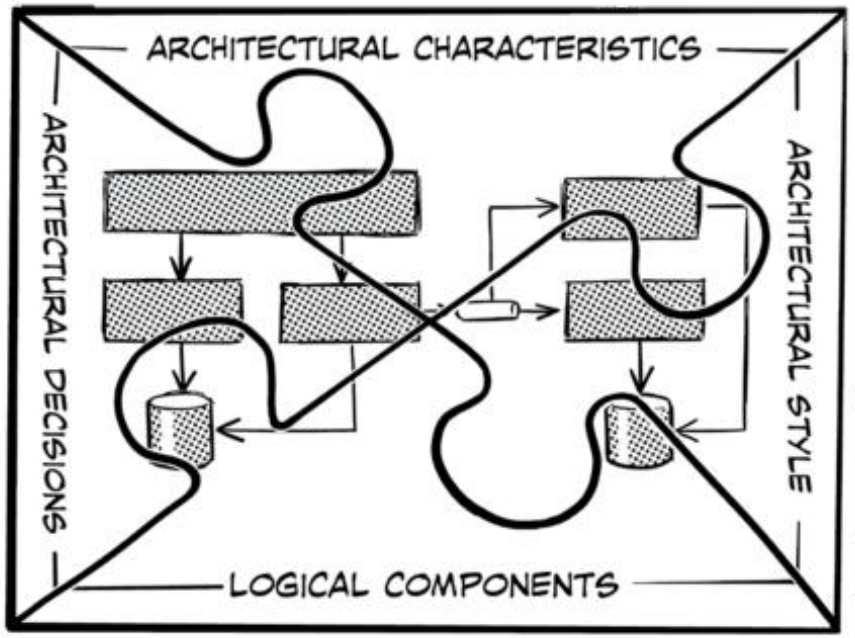


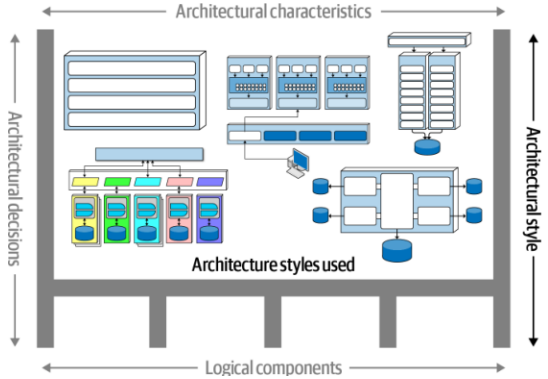
Leadership

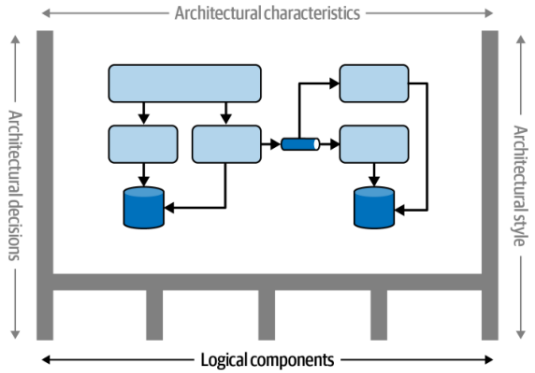
- Transparent
- Discipline
- Original


Architectural characteristics
Availability
Reliability
Testability
Scalability
Security
Agility
Fault Tolerance
Elasticity
Recoverability
Performance
Deployability
Learnability
Architectural decisions
Architectural style
Logical components


Architectural characteristics
Presentation component
Presentation layer
Closed
Business component
Business component
Business layer
Closed
Shared component
Services layer
Open
Persistence layer
Closed
Database layer
Closed
Only the Business and Services layers can access the Persistence layer
Architectural decisions
Architectural style
Logical components


ARCHITECTURAL CHARACTERISTICS
ARCHITECTURAL DECISIONS
ARCHITECTURAL STYLE
LOGICAL COMPONENTS


Architectural characteristics
Architectural decisions
Architectural style
Architecture styles used
Logical components


Architectural characteristics
Architectural decisions
Architectural style
Logical components

Universidad de La Sabana
FACULTAD DE INGENIERIA

Architectural Decision Record, ADR

- Transparent
- Discipline

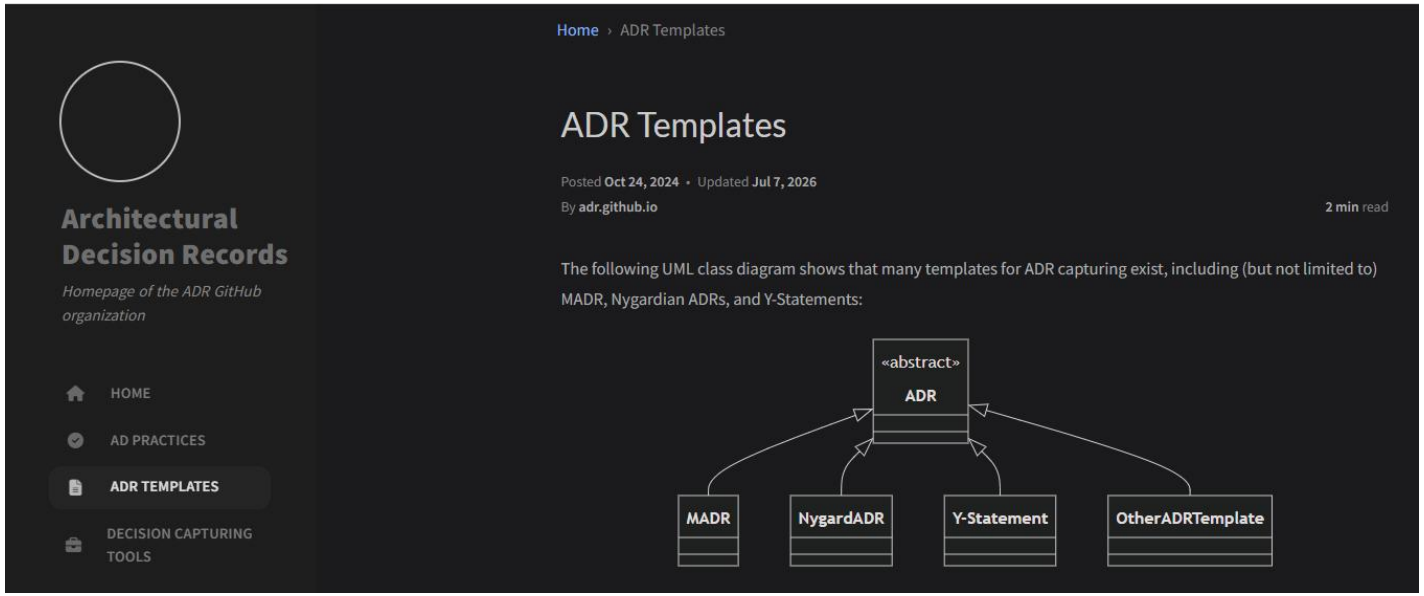
Why is important to keep a decision record ?

Hindsight bias (Juicio de la retrospectiva/*Sesgo del historiador*) criticize based on results only instead on the moment of taking the decision. (*Yo estaba alli…se seleccionó Salesforce por….*)

Document a decision based on:

- Needs/Requirements
- Context/Circumstances
- Consequences

Depends on depth, formality and consumers.



github.com/architecture-decision-record/architecture-decision-record

README | Code of conduct | License

Contents:

- What is an architecture decision record?
- How to start using ADRs
- How to start using ADRs with tools
- How to start using ADRs with git
- File name conventions for ADRs
- Suggestions for writing good ADRs
- ADR example templates
- Teamwork advice for ADRs
- Teamwork questions for ADRs
- Next step concepts for ADRs
- Architecture diagrams & views & viewpoints
- Fitness functions for decisions as code
- Decision guardrails for pull requests
- For more information

Templates:

- Decision record template by Jeff Tyree and Art Akerman
- Decision record template by Michael Nygard
- Decision record template by EdgeX
- Decision record template by arc42
- Decision record template for Alexandrian pattern
- Decision record template for business case
- Decision record template of the MADR project
- Decision record template using Planguage
- Decision record template by Paulo Merson

Programı

Contents:

- Summary
  - Issue
  - Decision
  - Status
- Details
  - Assumpti
  - Constrain
  - Positions
  - Argument
  - Implications
- Related
  - Related decisions
  - Related requirements
  - Related artifacts
  - Related principles
- Notes

Universidad de La Sabana
FACULTAD DE INGENIERIA

Culture eats strategy for breakfast

- Transparent

What is the competitive advantage of your company?

Is it helps to take decisions? No-obvious for a new hired.

Desmantelar la complejidad bancaria tradicional y democratizar el acceso financiero. → "Radical Simplicity & Humanize"



Dominar la logística e e-commerce ofreciendo la máxima velocidad. →"Frictionless Checkout & High Availability"

Are the ADRs on Arch Guidelines/ADKP/etc. are useful and updated, or a mandatory step ?

Is it can help for a fast decision? In an operative way? i.e. based on budgets (10k, 100k..5M)

Universidad de La Sabana
FACULTAD DE INGENIERIA

How you can evaluate it ?

- Discipline
- Original

Why do you evaluate?
Growth → Learning
Comparison → Employee Performance/Job Promotion/Provider-Service Selection (i.e. CRM every X years)
Pass a course…← Evaluation criteria, objective review



This isn't *the Spanish Inquisition.*

Level 1: Yourself (i.e. Tip about Interviews)
Level 2: Peer Review
Level 3: Outsiders (ARB or Consultancy/CSP), Objective Review?

Key Evaluation Activities

1. The reviewers individually ensure that they understand the current state of the architecture.
2. The reviewers determine a number of drivers to guide the review.
3. For each scenario, each reviewer should determine whether the scenario is satisfied
4. The reviewers capture potential problems exposed during the prior step.

- *The importance of the decision.* The
  be taken in making it and making sur
- *The number of potential alternatives.*
  spent in evaluating them.
- *Good enough as opposed to perfect.*

Universidad de La Sabana
FACULTAD DE INGENIERIA

ATAM and its adaptation

TABLE 21.2 ATAM Phases and Their Characteristics

Phase	Activity	Participants	Typical Cumulative Time
0	Partnership and preparation	Evaluation team leadership and key project decision makers	Proceeds informally as required, perhaps over a few weeks
1	Evaluation	Evaluation team and project decision makers	1–2 days
2	Evaluation (continued)	Evaluation team, project decision makers, and stakeholders	2 days
3	Follow-up	Evaluation team and evaluation client	1 week

Source: Adapted from [Clements 01b].

Ph1-2:
1: Present the ATAM
2: Present the Business Goals
3: Present the Architecture
4: Identify the Architectural Approaches
5: Generate a Quality Attribute Utility Tree
6: Analyze the Architectural Approaches

Ph2:
7: Brainstorm and Prioritize Scenarios
8: Analyze the Architectural Approaches
9: Present the Results

Outputs
- A concise presentation of the architecture
- Articulation of the business goals
- Prioritized quality attribute requirements expressed as quality attribute scenarios
- A set of risks and non-risks.
- A set of risk themes
- Mapping of architectural decisions to quality requirements.
- A set of identified sensitivity points and tradeoff points.

"Inception" (ARB)



SAFe BY SCALED AGILE

LeSS

3a: General Arch and goes deep.
3b: View by Subject Area: *Compliance, Networking, Security, DB, Storage, Operations, LZ –as implicit-, etc.*
4: Impact on running projects
--
7: Timeline/Epics/PI and Team
8a: Agreement between SMEs/Decision Makers
8b: Detected Tech Debt or out-of-scope

Other Options:
LAE – Lightweight Architecture Evaluation
Spike Review (Peer Review)

[SAiP], Ch. 21

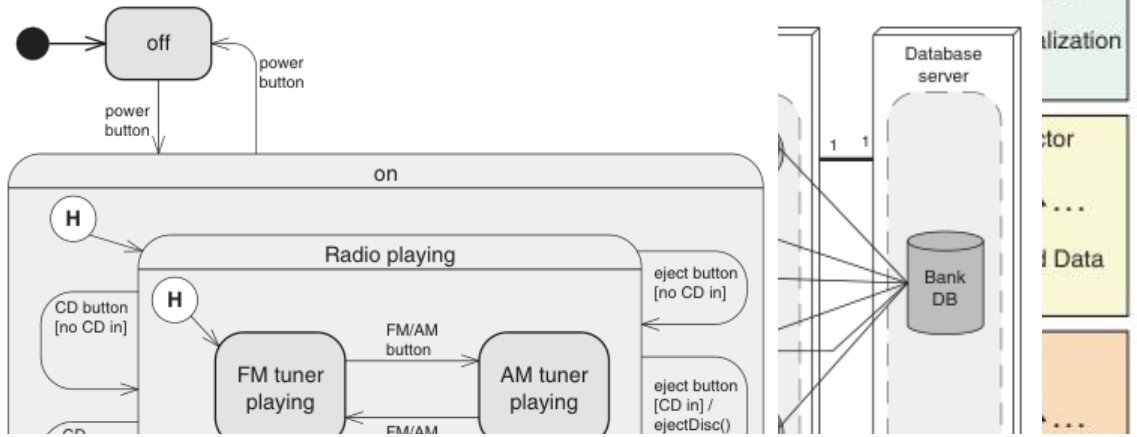
Universidad de La Sabana
FACULTAD DE INGENIERIA

How you can document it?

- Discipline
- Original

Why is important to keep a living documentation ? Arch doc serves..

- as a means of education.
- as a primary vehicle for communication among stakeholders.
- as the basis for system analysis and construction.
- as the basis for forensics when an incident occurs

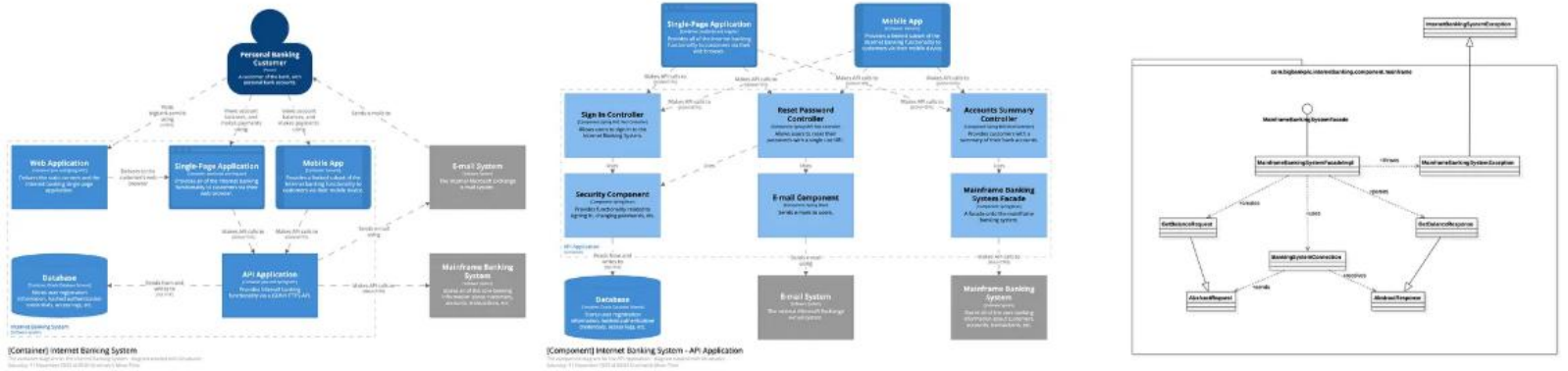


Notations:

Informal: Hand-written, mix of other types.
Semi-formal: UML, SysML.
Formal: Math semantic notation, ADL, ISO/IEC/IEEE 42010.

There are more options:
C4 Model, BPMN, ArchiMate, …

Depends on the platform scope, compliance, you have to go to Software Architecture Document (SAD).



C4 representations examples of Component | Container | Code

Universidad de La Sabana
FACULTAD DE INGENIERIA

Architectural Katas

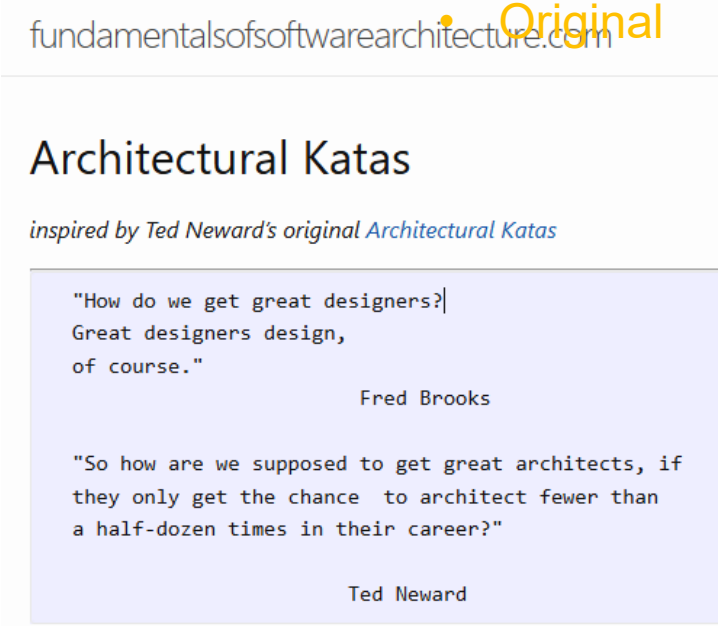
- Discipline
- Original

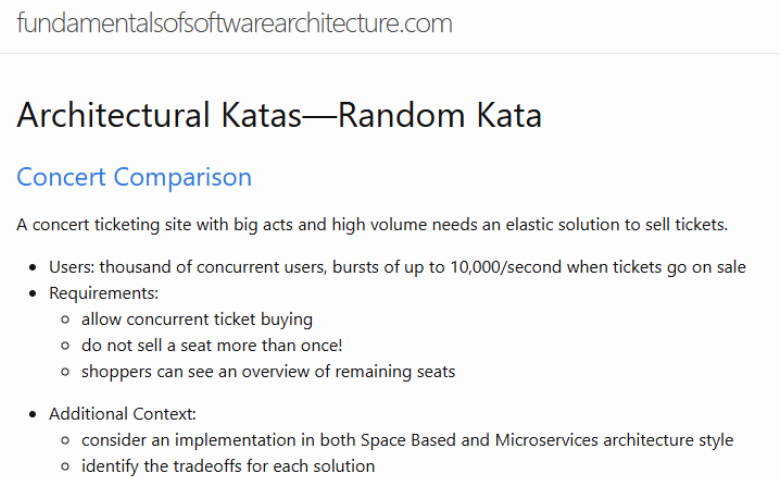
Gamification: Learning by playing

Pre-req:
Different (3-5 g) small groups (3-5 p).
Blackboard or virtual tools
-Opt- Previous project/RFP

Steps:

1. Give project/RFP and read it.
2. Questions to moderator.
3. Stablish rules: Min. Diagram (C2/C4, Deploy UML)
4. -Timing-
5. Explain to the other groups: Hard-but-fair questions, please **don't escalate to moderator.**
6. Voting
7. Next group…

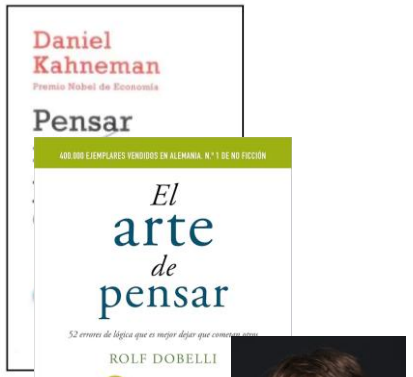


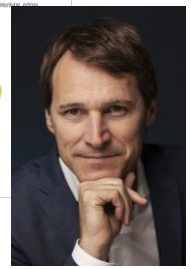


Universidad de La Sabana
FACULTAD DE INGENIERIA

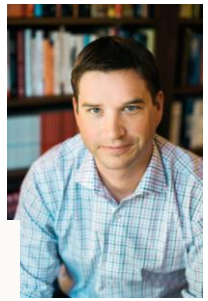
Keep update...

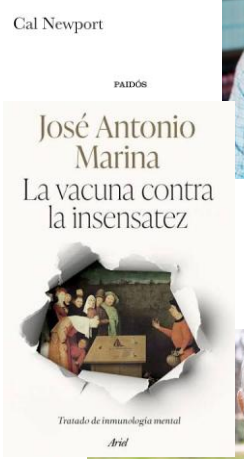
- Discipline
- Original





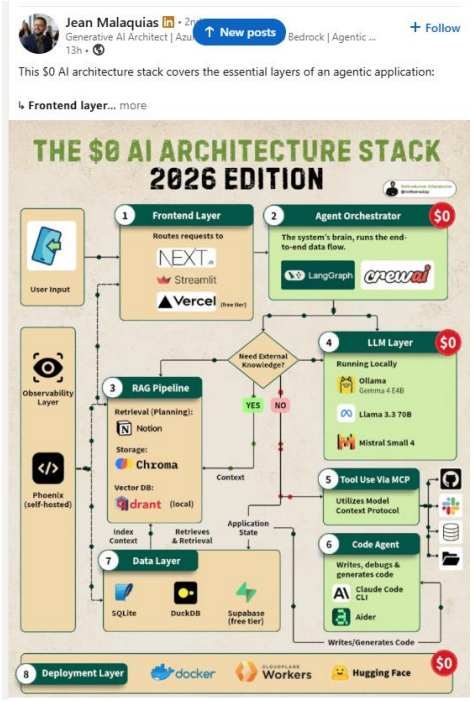


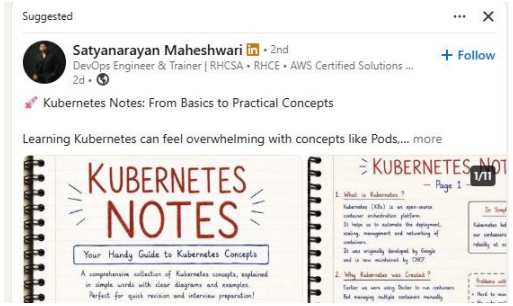






It's your time…





Tech Attendance

Speaker



Universidad de La Sabana
FACULTAD DE INGENIERIA

Keep update...

- Discipline
- Original

Am I a follower ?

User Groups:
AWS, GCP, etc. I started with locals.
University, Focus Groups.

CSP' VP, Managers, etc.:

Am I a learner ?

Evangelists:

Educational:

Blogs:

Am I a reader ?

Andrej Karpathy

Gregor Hohpe · 2nd
Redefining the role of architects. Retired from big tech. Not retired from trying to make IT a better place.

Dr Milan Milanović · Following
Helping 400K+ engineers and leaders grow

Martin Fowler · 3rd
Author of Refactoring, hosts martinfowler.com, Thoughtworker
Boston, Massachusetts, United States · Contact info

Sam Newman · Following
Independent technology consultant, author of Building Microservices and ..
6h •

Adrian Cockcroft · 2nd
Tech Advisor
Greater Salinas Area · Contact info

Luca Mezzalira · 2nd
Architecture in the Age of AI · Dis

Andrew Ng · 3rd
DeepLearning.AI, AI Fund and AI Aspire
Palo Alto, California, United States · Contact info

Corey Quinn · 2nd
Writes Artificial Confidence, the AI expository newsletter your CTO forwards you • Cloud Economist, Duckbill
San Francisco, California, United States · Contact info

Simon Sinek · Following
Optimist, New York Times bestsel
53m • Edited •

Sander Van Vugt · 1st
Independent author and trainer on Linux, Ansible and Kubernetes. NOT in the market for a job but available as trainer and consultant.

Yury Niño Roa · 1st
Google Cloud Application Engineer | SRE & DevOps & Chaos Engineering Advocate

Anabel Montero Posada · 1st
AWS Technical Trainer | Data lover | Cloud Enthusiast
Bogota, D.C., Capital District, Colombia · Contact info

CÉSAR Mendoza Canales · 1st
AWS Senior Technical Trainer @ Amazon Web Services | Agentic AI Architect | Enterprise GenAI Enablement (LATAM) | MIT & Cambridge CTO Programs
Mexico City, Mexico · Contact info

Dorie Clark · Following
WSJ & USA Today Bestselling Au
View my newsletter
1h •

Carlos Eduardo Ortiz · 1st
Solution Architect Cloud en Oracle
Bogota, D.C., Capital District, Colombia · Contact info

Kate Udalova · 2nd
Founder of 7taps | MicrolearningCONF | Stanford instructor
Dubai, United Arab Emirates · Contact info

CLAUDERS CHAPTER — COLOMBIA
Bogota
LAB101 UNAL
37m •

Jeff Barr · 2nd
Vice President & Chief Evangelist at Amaz

Priyanka Vergadia · 2nd
#1 Visual Storyteller in Tech | VP Level Product & GTM | TED Speaker | Enterprise AI Adoption at Scale | 250K+ Community
Mountain View, California, United States · Contact info

Frank Kane · 2nd
Teaching AI and tech skills to over 1M worldwide
Los Angeles Metropolitan Area · Contact info

Python Developers Community (moderated)
Public group

KodeKloud
23m •

Stéphane Maarek · Following
3M+ students trained on Udemy | 10x AWS

Adrian Cantrill · 2nd
AWS/Cloud Course Creator @ learn.cantrill.io // follow for updates! connect to chat // youtube.com/c/LearnCantrill
Brisbane, Queensland, Australia · Contact info

IPSpecialist
The Cloud. Simplified. Secured. By Experts you can trust

Jon Bonso · 2nd

Neal K. Davis · 2nd
I help career changers and tech professionals build hands-on AWS AI skills to land high-paying cloud jobs with real-world training

Rag Dhiman · 2nd
Technical Instructor | Pluralsight Author x20+ | Linkedin Instructor | Author | CEO | AI Engineer

K21Academy: Learn AI, Data & Cloud from Experts
Transition into IT → Certified in 6 Months | Land High-Paying Roles in AI, Data & C
Success Stories
Information Technology & Services · London · 14K followers · 51-200 employees

Oracle Developers
1h •

Last week in AWS

Universidad de La Sabana
FACULTAD DE INGENIERIA

Soft Skills

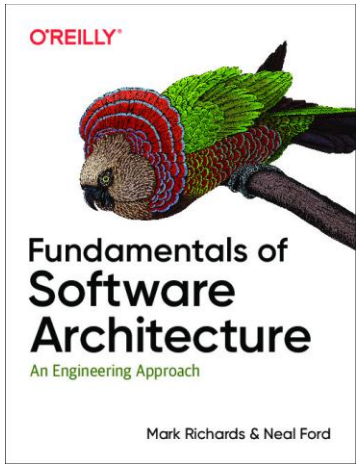
- Original

Written comms.

- Part III. Techniques and Soft Skills
  - Chapter 21. Architectural Decisions
    - Architectural Decision Antipatterns
    - Architectural Significance
    - Architectural Decision Records
  - Chapter 22. Analyzing Architecture Risk
    - Risk Matrix
    - Risk Assessments
    - Risk Storming
    - User-Story Risk Analysis
    - Risk-Storming Use Case
    - Summary
  - Chapter 23. Diagramming Architecture
    - Diagramming
    - Summary
  - Chapter 24. Making Teams Effective
    - Collaboration
    - Constraints and Boundaries
    - Architect Personalities
    - How Much Involvement?
    - Team Warning Signs
    - Leveraging Checklists
    - Providing Guidance
    - Summary
  - Chapter 25. Negotiation and Leadership Skills
    - Negotiation and Facilitation
    - The Software Architect as a Leader
    - Integrating with the Development Team
    - Summary
  - Chapter 26. Architectural Intersections
    - Architecture and Implementation
    - Architecture and Infrastructure
    - Architecture and Data Topologies
    - Architecture and Engineering Practices
    - Architecture and Team Topologies
    - Architecture and Systems Integration
    - Architecture and the Enterprise
    - Architecture and the Business Environment
    - Architecture and Generative AI
    - Summary
  - Chapter 27. The Laws of Software Architecture, Revisited
    - First Law: Everything in Software Architecture Is a Trade-Off
    - Second Law: Why Is More Important Than How
    - The Spectrum Between Extremes
    - Parting Words of Advice

- Architect Personalities
  - The Control-Freak Architect
  - The Armchair Architect
  - The Effective Architect

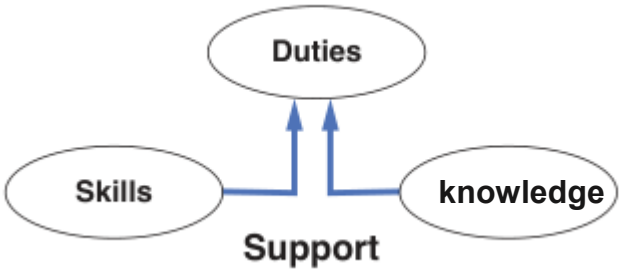
Interpersonal comms.



Universidad de La Sabana
FACULTAD DE INGENIERIA

Soft Skills

- Original

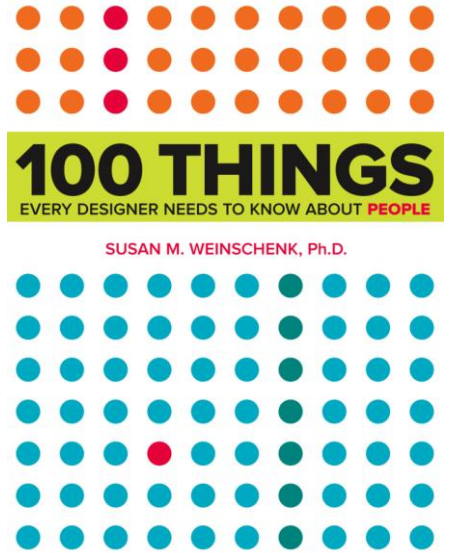


Gain experience carrying out the duties.

Apprenticeship is a productive path to achieving experience. Education alone is not enough, because education without on-the-job application merely enhances knowledge.

Improve your nontechnical skills.

Master the body of knowledge



	Performing other architecting duties	Sell the vision. Keep the vision alive. Participate in product design meetings. Give technical advice on architecture, design, and development. Provide architectural guidelines for software design activities. Lead architecture improvement activities. Participate in software process definition and improvement. Provide architecture oversight of software development activities.
Duties concerned with life-cycle activities other than architecting	Managing the requirements	
	Evaluating future technologies	
	Selecting tools and technology	
	Supporting the business	Understand and evaluate business processes. Translate business strategy into technical strategy. Influence the business strategy. Understand and communicate the business value of software architecture. Help the organization meet its business goals. Understand customer and market trends.
	Managing the people on the architect's team	Build "trusted advisor" relationships. Coordinate. Motivate. Advocate. Train. Act as a supervisor. Allocate responsibilities.

Communication skills	Outward communication (beyond the team)
	Inward communication (within the team)
Interpersonal skills	Team relationships

Knowledge of architecture concepts

Knowledge of software engineering

Design knowledge

Programming knowledge

Specific technologies and platforms

General knowledge of technologies and platforms

Domain knowledge

Industry knowledge

Business knowledge

Leadership and management techniques

Soft Skills

Why do you don’t show your PoC/PoV ?

Test yourself….what is your exposition ?

Do you use an accurate language ?

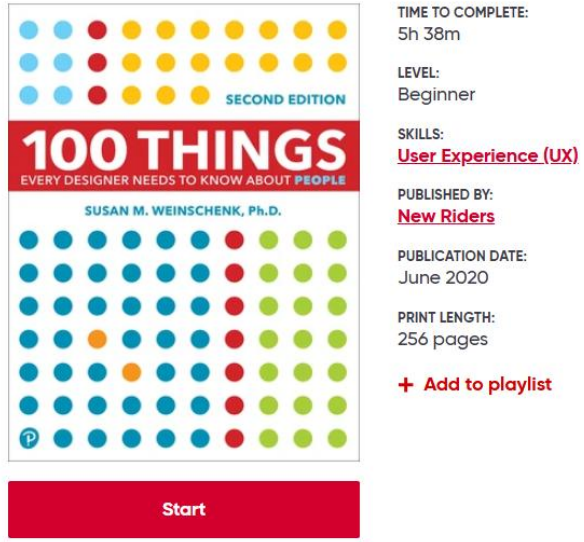
Does your thoughts are out-of-the-box ?

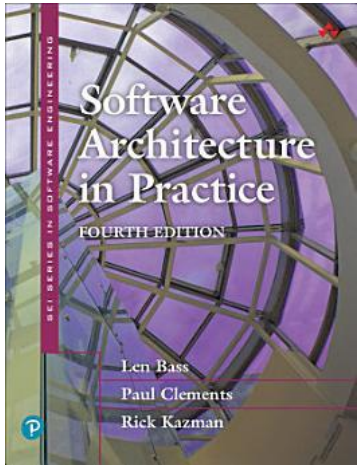
Can you explain your position and your interests easily ?

Universidad de La Sabana
FACULTAD DE INGENIERIA

References

- [ADR] Nygard, M., & Koppelman, J. (n.d.). ADR GitHub Project. https://adr.github.io/
- [SAiP] Bass, L., Clements, P., & Kazman, R. (2021). Software architecture in practice (4th ed.). Addison-Wesley Professional. https://learning-oreilly-com.ez.unisabana.edu.co/library/view/software-architecture-in/9780136885979/
- [FoSA] Richards, M., & Ford, N. (2025). Fundamentals of software architecture (2nd ed.). O'Reilly Media. https://learning-oreilly-com.ez.unisabana.edu.co/library/view/fundamentals-of-software/9781098175504/,
- [AKA] Richards, M., & Ford, N. (n.d.). Software Architecture Katas. https://fundamentalsofsoftwarearchitecture.com/katas/
- Amazon Web Services. (n.d.). AWS Architecture Center. https://aws.amazon.com/architecture/
- Amazon Web Services. (n.d.-b). AWS Prescriptive Guidance. https://aws.amazon.com/prescriptive-guidance/
- Microsoft. (n.d.). Azure Architecture Center. Microsoft Learn. https://learn.microsoft.com/en-us/azure/architecture/
- Microsoft. (n.d.). Browse Azure Architectures. Microsoft Learn. https://learn.microsoft.com/en-us/azure/architecture/browse/
- Oracle. (n.d.-a). Cloud Architecture Center. https://www.oracle.com/cloud/architecture-center/
- Oracle. (n.d.). Oracle Architecture Center: Reference Architectures. https://docs.oracle.com/solutions/?q=&cType=reference-architectures&sort=date-desc&lang=en
- Google. (n.d.). Cloud Architecture Center. https://docs.cloud.google.com/architecture
- Google. (n.d.). Google Cloud solutions. https://cloud.google.com/solutions?hl=en





Universidad de La Sabana

FACULTAD DE INGENIERÍA

CONTACTO

Francisco Moreno

Profesor Catedra – Facultad de Ingeniería

franciscomodi@unisabana.edu.co